

TITANIC DISASTER: EXPLORATORY DATA ANALYSIS REPORT

Data Science Portfolio Practice — Statistical & Visual Exploration

1. Executive Summary & Data Integrity Audit

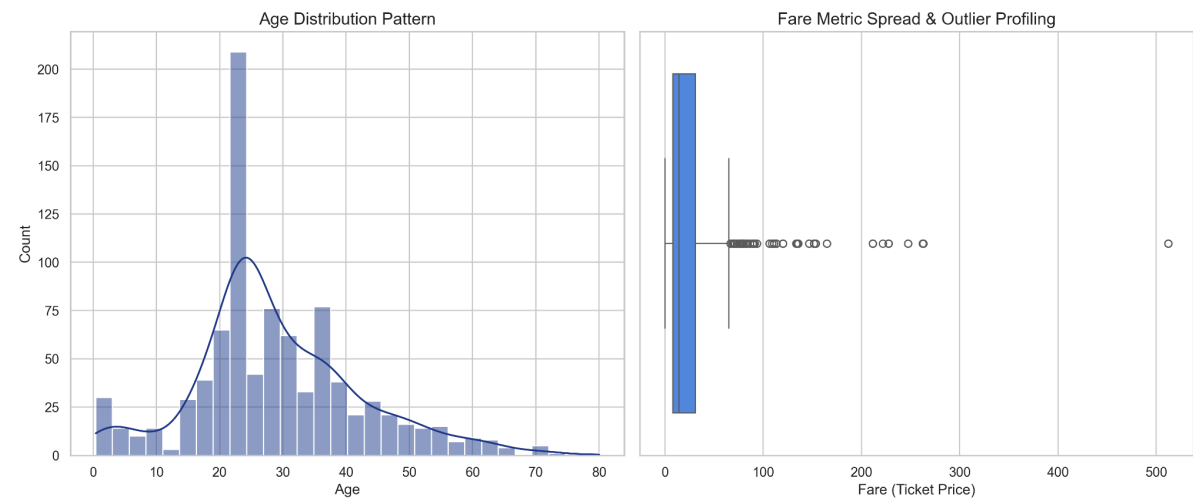
This comprehensive analytical report evaluates the historical passenger manifest records of the RMS Titanic (891 rows, 12 core features). The purpose of this analysis is to visually and statistically explore demographic trends, pricing structures, and socio-economic variables to isolate the primary indicators of passenger survival probability.

An initial data integrity check revealed structural missingness across key columns, which were handled prior to visualization to ensure unbiased insights:

- **Age Vector:** 177 missing inputs (19.9% missingness). Repaired using stratified median values across Passenger Classes (Pclass) to maintain demographic variance.
- **Cabin Identifier:** 687 missing inputs (77.1% missingness). Due to the severe data gaps, this attribute was excluded from downstream analysis blocks.
- **Embarked Port:** 2 missing inputs. Imputed using the global column mode ("S" for Southampton).

2. Univariate Analysis: Feature Distributions & Outliers

Statistical distribution analysis shows a near-normal age distribution across the ship's demographic footprint, peaking within the 20 to 38 chronological range. Conversely, financial ticket fares display an extreme right-skewed layout (0.00 to 512.33), highlighting heavily isolated luxury anomalies on upper-deck classes.

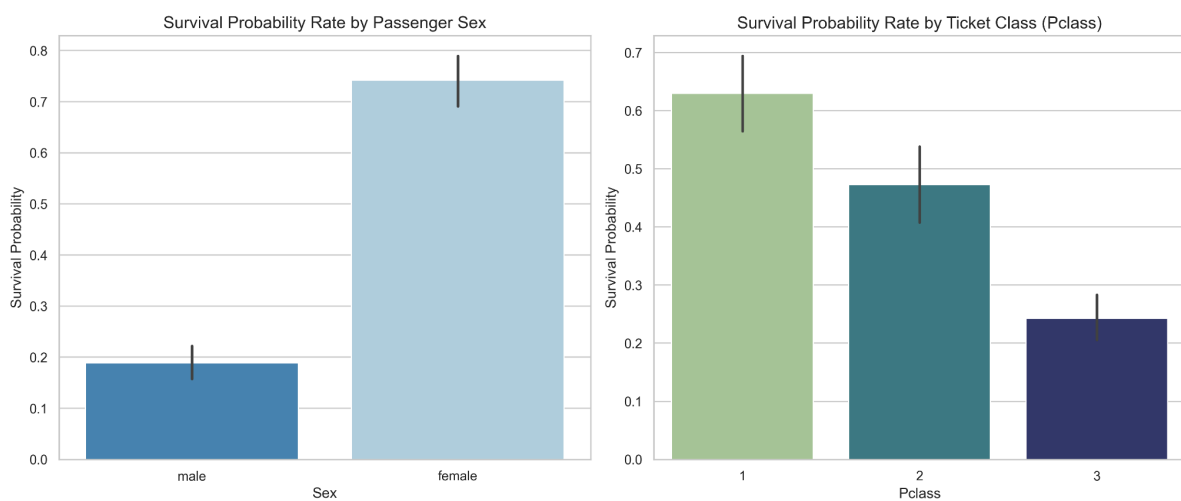


Key Visual Observations:

- **Age Distribution:** The distribution confirms that the vast majority of passengers were young working-class individuals and families in their prime. The distribution trails off naturally on both ends (infants and the elderly).
- **Fare Outlier Profiling:** The boxplot exposes a heavily skewed pricing structure. While 75% of passengers paid less than 31.00, a handful of elite tickets spiked over 200, with an extreme maximum at 512.33. This severe variance ($\text{std} = 49.69$) requires log-transformations or standard scaling adjustments before training linear estimators.

3. Bivariate Analysis: Target Survival Correlates

Evaluating categorical splits against survival flags reveals deep systemic socio-demographic stratification factors during emergency evacuations.

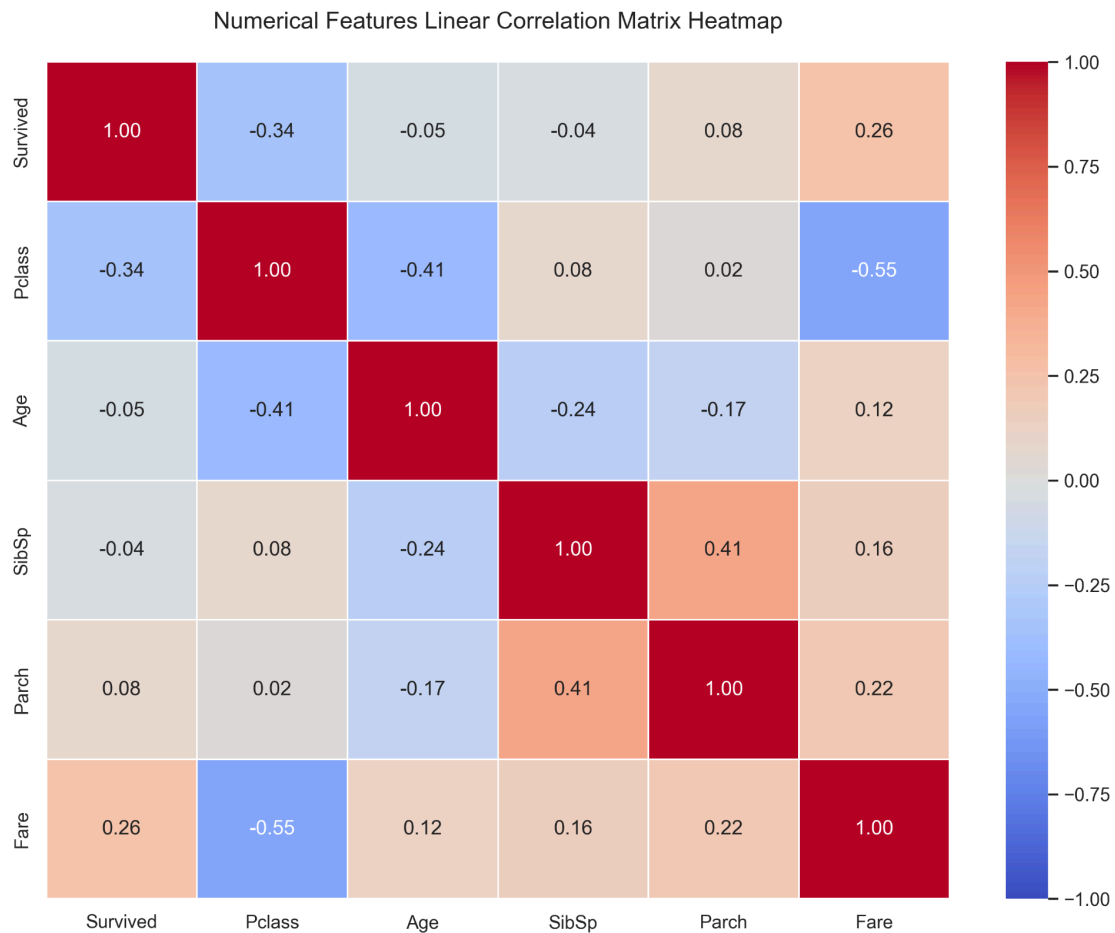


Key Visual Observations:

- **Gender Demographics (The Chivalry Effect):** Females achieved a **74.2%** survival probability, contrasted against a minor **18.9%** rate for male counterparts. This strongly quantifies the profound impact of the historical "women and children first" maritime evacuation protocol.
- **Socioeconomic Class (The Wealth Gap):** Ticket stratification analysis uncovers a clear disparity across socio-economic tiers. Upper-class travelers (Pclass 1) secured a **63.0%** survival rate, whereas lower-class travelers (Pclass 3) dropped to **24.2%**, signaling restrictive lower-deck evacuation access during the event.

4. Multivariate Analysis: Pearson Linear Interaction Matrix

Isolating numerical parameters allows us to identify multicollinearity issues and underlying connections across tracking features.



Key Visual Observations:

- **Pclass vs. Fare Correlation:** A strong negative correlation coefficient of **-0.55** exists between Passenger Class (Pclass) and Ticket Fare, confirming pricing segmentation (as class number drops from 3rd to 1st, fare rises sharply).
- **Age vs. Pclass Interaction:** A slight inverse relationship reveals that older, more financially established passengers disproportionately populated the premium upper-deck cabins.
- **Multicollinearity Warning:** The strong linear bond between ticket price and passenger tier indicates that multicollinearity must be tracked carefully using Variance Inflation Factors (VIF) prior to feature selection in machine learning pipelines.